Eugene Belilovsky

Academic Work Experience

Concordia University, Montreal, Canada *Dept. of Computer Science and Software Engineering* (November 2020 - Present)

- Associate Professor (Tenured), July 2025-Present
- Assistant Professor, November 2020 - June 2025
- Adjunct Professor at University of Montreal, DIRO, 2020 - Present
- Associate Academic Member at Mila (Quebec Institute of Artificial Intelligence), 2020 - Present

University of Montreal & Mila , Montreal, Canada *Postdoctoral Researcher* (July 2018-Nov 2020)

- Supervised by Aaron Courville
- Topics: Continual Learning, Visual Scene Understanding, Embodied Agents, and Distributed training
- Published in ICML, NeurIPS, ECCV, BMVC, JMLR, TPAMI.

University of Toronto, Toronto, Canada *Visiting PhD Researcher* (Summer 2016)

- Supervised by Richard Zemel and Raquel Urtasun
- Topic: Multi-modal representation learning

Industry Work Experience

Amazon Development Center, Berlin, Germany *PhD Intern* (Winter 2018)

Apple Inc, Cupertino, California, USA *PhD Intern* (Summer 2017)

International Biometrics Group, New York City, USA *Computer Vision Research Engineer* (July 2011-December 2013)

ITT Corporation, Clifton, USA *Embedded Systems Software Engineer* (August 2010 - July 2011)

Education

CentraleSupelec, University of Paris-Saclay Gif-Sur-Yvette, France

- PhD, March 2014 - March 2018
- Advisor: Matthew Blaschko
- Thesis: Structured Sparse Learning on Graphs in High-Dimensional Data with Applications to Neuroimaging

KU Leuven (Co-tutelle) Leuven, Belgium

- PhD, Electrical Engineering, March 2015 - March 2018
- Degree based on work completed at both KU Leuven and CentraleSupelec

Cooper Union New York City, NY, U.S.A.

- Bachelors of Engineering (Summa Cum Laude), Electrical Engineering, May 2010
- Masters of Engineering, Electrical Engineering, May 2011

Organization

- Co-organizer of ICML 2023 Workshop: Localized Learning: Decentralized Model Updates via Non-Global Objectives
- Co-organizer of ICML 2020 Workshop: Continual Learning
- Co-organizer of ICLR 2019 Workshop: Learning with Limited Labeled Data: Weak Supervision and Beyond
- Co-organizer of NeurIPS 2017 Workshop: Learning with Limited Labeled Data: Weak Supervision and Beyond

Academic Service

- Mila EDI Committee 2024-2025
- Area Chair: NeurIPS 2025, ICLR 2025-2026, BMVC 2023
- Reviewer at NeurIPS 2016-2024, ICML 2017-Present, ICLR 2017-2024, CVPR 2022-2023, WACV 2025
- Ad-hoc reviewer at TPAMI and Computerized Medical Imaging
- Member of 20+ PhD and MS committees (2020-Present)
- Member IVADO PhD Scholarship Committee 2022
- Member FRQNT Scholarship (Master's) Committee 2022-2025
- Member Concordia CSSE Department Award Committee 2023-2025
- Subject Matter Expert for Committe Developing BSc in AI at Concordia 2025
- Expert reviewer for Marsden Fund Grant, New Zealand 2023

Teaching

- Developed and Taught Graduate and Undergraduate Deep Learning Course at Concordia University 2021-2024
- Developed and Taught Federated Learning Course at Concordia University 2023-Present
- Undergraduate Control Systems 2022-2023

Patents

- EP24178061.8 w. Toyota ”Controlling Forgetting with Test-Time Data in Continual Learning”
- EP23162437.0 w. Toyota ”Prototype-Sample Relation Distillation: Towards Replay-Free Continual Learning”

Software

- Co-developer of kymatio (scattering software): kymat.io
- Co-developer for PyLO software package: pylo.readthedocs.io

Supervision

Currently Supervising: 1 PostDoc, 8 PhD, 3 MSc

- PostDoc: Geraldin Nanfack - **Concordia University** (January 2023 - Present) , NSERC PostDoc Award
- PhD Student: Abhinav Moudgil - **Concordia University** (September 2021 - Present) FRQNT PhD Award
- PhD Student Gwendolyne Legate - **Concordia University** (January 2023- Present) FRQNT PhD, Borealis AI Award
- PhD Student: Mohammadreza Davari - **Concordia University** (January 2020 - Present)
- PhD Student: Albert Orozco Camacho - **Concordia University** (September 2022 - Present)
- PhD Student: Adel Nabli - **Concordia University and Sorbonne (Co-tutelle)** (January 2022 - Present)
- PhD Student: Benjamin Therien - **University of Montreal** (September 2023 - Present) FRQNT PhD Award
- PhD Student: Vaibhav Singh - **Concordia University** (September 2023 - Present)
- PhD Student: Amirhossein Zamani - **Concordia University** (January 2024 - Present)
- PhD Student: Paul Janson - **Concordia University** (September 2023 - Present) Fast-Tracked
- MS Student: Paria Mehrbod - **Concordia University** (September 2023 - Present)
- MS Student: Zafir Khalid - **Concordia University** (May 2024 - Present)
- MS Student: Xiaolong Huang - **Concordia University** (January 2025 - Present)

Completed Supervision: 1PhD (Sole supervised), 13 MSc Thesis (6 Sole supervised), 3 PhD Interns, 4 Undergrad NSERC URA

- MS Student: Nicolas Bernier - **Concordia University** (September 2022 - Present)
  Now Research Engineer at Eleven Labs
- MS Student: Humza Wajid - **Concordia University** (January 2023 - Present)
  Now Lead Software Engineer at Majesco
- MS Student: Charles-Etienne Joseph - **University of Montreal** (January 2023 - May 2024)
  Now Research Engineer at Capital One, Pretraining Team.
- PhD Student: Mohammadreza Davari - **Concordia University** (January 2021 - July 2025)
  Now Senior Applied Scientist at Microsoft
- MS Student: Congshu Zou - **Concordia University** (September 2023 - Present)
- MS Student: Alexander Fulleringer - **Concordia University** (September 2022 - Present)
  Now Instructor at CEGEP
- MS Student: Amir Sarfi - **Concordia University** (January 2022 - March 2023)
  Now Research Scientist at Covenant AI
- MS Student: Muawiz Chaudhary - **Concordia University** (September 2021 - July 2023)
  Now Founding Research Engineer at Guidelabs.ai
- MS Student Irene Tenison (co-supervised with Irina Rish) - **University of Montreal** (August 2021 - December 2022)
  Now PhD Student at MIT with Tim Berners-Lee
- MS Student Gwendolyne Legate - **University of Montreal** (February 2022 - December 2022)
  Continued to PhD
- MS Student Medric Sonwa - **University of Montreal** (January 2022 - January 2023)
  Now Senior AI Engineer at Deloitte
- MS Student Nader Asadi - **Concordia University** (July 2021 - April 2023) w. Sudhir Mudur
  Now Research Scientist at Haiper AI
- MS Student: Adeetya Patel - **Concordia University** (September 2021 - March 2023)
  Now Data Scientist at McGill
- MS Student: Amir Sarfi - **Concordia University** (September 2021 - March 2023)
  Now Senior Researcher at Covenant AI
- MS Student: Nasir Khalid Mohammad - **Concordia University** (May 2021 - March 2023) w. Tiberiu Popa
  Now Senior Member of Technical Staff Runway
- NSERC Undergraduate RA: William Chittavong - **Concordia University** (May 2024 - September 2024)
- NSERC Undergraduate RA: Eli Samuel - **Concordia University** (May 2023 - September 2023)
  Now Research Software Engineer at Silicolabs
- NSERC Undergraduate RA: Yixiuan Xang - **Concordia University** (May 2022 - September 2022)
  Now Software Engineer at Amazon
- NSERC Undergraduate RA: Benjamin Therien - **Concordia University** (May 2021 - September 2021)
  Continued to PhD
- Co-supervisor of PhD Intern Boris Knyazev - **Mila** (October 2019 - May 2020)
  Now Research Scientist at Samsung
- Co-Supervised Master’s thesis of Thomas Strypsteen - **KU Leuven** (Fall 2017 - Spring 2018)

Research Funding

- OpenPhilanthropy 2026-2028 (PI, **210K** USD)
- Google-Mila 2025-2026 (PI, **50K** CAD)
- Samsung-Mila 2024-2025(**65K** CAD)
- MITACS Service Now 2025 (PI, **32K** CAD)
- NFRF Exploration 2025-2027(co-PI with Yiming Xiao, **200K CAD** CAD)
- NSERC Alliance with Toyota Research (PI, **160K** CAD)
- NSERC Discovery 2021-2026 (PI, **132K** CAD)
- OpenPhilanthropy 2023-2025 (PI, **131K** CAD)
- CFI John Evans Leader Fund 2023-2027 (PI with Mirco Ravanelli, **473K** CAD)
- Concordia Startup-Fund 2020-2023 (PI, **65K** CAD)
- Concordia Facility Optimization Award 2022-2023 (PI, **35K** CAD)
- Digital Research Alliance RRG Competition 2025-2026 - (PI, **396K** CAD, Year 1 eligible for fast-track)
- Digital Research Alliance RRG Competition 2022-2025 - (PI, **185** CAD)
- IVADO PRF3 2023-2024 (co-PI, share **59K** CAD)
- FRQNT New Scholar 2023-2025 (PI, **60K** CAD)
- FRQNT-NOVA 2023-2026 (co-PI with Guy Wolf, Jian Tang, Danica Sutherland; **225K** CAD)
- FRQNT-NOVA 2023-2026 (co-PI with Narges Armanfard, Graham Taylor, Samira Kahou; **225K** CAD)
- Mila-Tech Transfer Grant with Silicolabs 2024-2025 (co-PI with Guillaume Dumas **141K**)

Honors and Awards

- IVADO Postdoctoral Fellowship 2019-2021 (140,000 CAD)
- MITACS-INRIA Globallink Award (Mobility) 2016 (5000 CAD)
- Mobility Grant from STIC doctoral school 2016 (1000 euro)
- University of Paris Saclay STIC best contribution award (2nd place) 2016
- William C. and Esther Hoffman Beller Prize for merit in engineering studies 2010
- DAAD RISE Research Scholarship 2009
- Full Tuition Scholarship at the Cooper Union 2006-2011
- Cooper Union Dean's List 2006-2010

Publications and Conference Presentations

Student contributions denoted with *
h-index: 30, Citations: 4229, 31 papers at A* conferences

Journal Publications

(J1) A. Moudgil*, B. Knyazev, G. Lajoie, **E. Belilovsky**. Celo: Training Versatile Learned Optimizers on a Compute Diet. *Transactions on Machine Learning Research (TMLR)* 2025. Journal-to-Conference at ICLR 2026 (10% of accepted papers).

(J2) B. Thérien*, C.-É. Joseph*, Z. Sarwar, A. Panda, A. Das, S.-X. Zhang, S. Rawls, S. Sahu, **E. Belilovsky**, I. Rish. Continual Pre-training of MoEs: How robust is your router? *Transactions on Machine Learning Research (TMLR)* 2025.

(J3) J. Stanley, E. Rabot, S. Reddy, **E. Belilovsky**, L. Mottron, D. Bzdok. Large language models deconstruct the clinical intuition behind diagnosing autism. *Cell* 2025. Journal impact Factor is 42.5

(J4) Y. Qi, P. Vianna, A. Cadrin-Chênevert, K. Blanchet, E. Montagnon, **E. Belilovsky**, G. Wolf, L.-A. Mullie, G. Cloutier, M. Chassé, A. Tang. Simulating Federated Learning for Steatosis Detection Using Ultrasound Images. *Scientific Reports*, 2024.

(J5) A. Ibrahim*, B. Thérien*, K. Gupta*, M. L. Richter, Q. Anthony, T. Lesort, **E. Belilovsky**, I. Rish . Simple and Scalable Strategies to Continually Pre-train Large Language Models. *Transactions on Machine Learning Research (TMLR)* 2024.

(J6) I. Tenison*, S. Sreeramadas, , V. Mugunthan, E. Oyallon, I. Rish, **E. Belilovsky**. Gradient Masked Averaging for Federated Learning *Transactions on Machine Learning Research (TMLR)* 2023.

(J7) P. Vianna*, S. Calce, P. Boustros, C. Larocque-Rigney, L. Patry-Beaudoin, Y. Luo, E. Aslan, J. Marinos, T. Alamri, K. Vu, J. Murphy-Lavallée, J. Billiard, E. Montagnon, H. Li, S. Kadoury, B. Nguyen, S. Gauthier,B. Therien*, I. Rish, **E. Belilovsky**, G. Wolf, M. Chassé, G. Cloutier, A. Tang Comparison of Radiologists and Deep Learning for US Grading of Hepatic Steatosis. *Journal of Radiology*. vol. 309, 2023. Journal impact factor 29.1

(J8) M. Andreux, T. Angles, G. Exarchakis, R. Leonarduzzi, G. Rochette, L. Thiry, J. Zarka, S.Mallat, J. andén, **E. Belilovsky**, J. Bruna, V. Lostanlen, M.J. Hirn, E. Oyallon, S. Zhang, C. Cella, M. Eickenberg. Kymatio: Scattering Transforms in Python. *Journal of Machine Learning Research (JMLR)* vol. 21 pp. 1-6, 2020. Journal impact factor 4.1

(J9) E. Oyallon, S. Zagoruyko, G. Huang, N. Komodakis, S. Lacoste-Julien, M. Blaschko, **E. Belilovsky**. Scattering Networks for Hybrid Representation Learning. *IEEE transactions on pattern analysis and machine intelligence (TPAMI)* vol. 41 pp. 2208 - 2221, 2018. Journal impact factor 20.8.

(J10) **E. Belilovsky**, A. Argyriou, G. Varoquaux, and M. B. Blaschko: Convex Relaxations of Penalties for Sparse Correlated Variables With Bounded Total Variation. Springer Machine Learning Journal, vol. 100, pg 533-553 2015. *ECML Journal Track*. Journal impact factor 7.5

(J11) **E. Belilovsky** et al: Predictive sparse modeling of fMRI data for improved classification, regression, and visualization using the k-support norm. *Computerized Medical Imaging and Graphics Journal*, vol. 46, pp. 40-46,2015. Journal impact factor 5.4

Conference Proceedings

(C1) P. Mehrbod, P. Vianna, G. Nanfack, G. Wolf, **E. Belilovsky**. Test Time Adaptation Using Adaptive Quantile Recalibration. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026.

(C2) A. H. Zamani, T. Xie, A. G. Aghdam, T. Popa, **E. Belilovsky**. Geometry-Aware Preference Learning for 3D Texture Generation. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026.

(C3) T. Xie, N. Aigerman, **E. Belilovsky**, T. Popa. Sketch-guided cage-based 3D Gaussian Splatting Deformation. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026.

(C4) A. Nabli*, L. Fournier, P. Erbacher, L. Serrano, **E. Belilovsky**, E. Oyallon. ACCO: Accumulate while you Communicate, Hiding Communications of Sharded Distributed LLM Training. *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

(C5) B. Knyazev, A. Moudgil*, G. Lajoie, **E. Belilovsky**, S. Lacoste-Julien. Accelerating Training with Neuron Interaction and Nowcasting Networks. *International Conference on Representation Learning (ICLR)* 2025.

(C6) D. Martins Gomes*, Y. Zhang, **E. Belilovsky**, G. Wolf, M. Hosseini. AdaFisher: Adaptive Second Order Optimization via Fisher Information. *International Conference on Representation Learning (ICLR)* 2025.

(C7) S. Rivaud, L. Fournier*, T. Pumir, **E. Belilovsky**, M. Eickenberg, E. Oyallon. PETRA: Parallel End-to-End Training with Reversible Architectures. *International Conference on Representation Learning (ICLR)* 2025.

(C8) R. Davari*, **E. Belilovsky**. Model Breadcrumbs: Scaling Multi-Task Model Merging with Sparse Masks. *European Conference on Computer Vision (ECCV)*, Milan, Italy, 2024.

(C9) S. Horoi*, A.M. Orozco Camacho*, **E. Belilovsky**, G. Wolf. Harmony in diversity: Merging neural networks with canonical correlation analysis. *International Conference on Machine Learning (ICML)*, Vienna, Austria, 2024.

(C10) P. Vianna*, M. Chaudhary*, P. Mehrbod*, A. Tang, G. Cloutier, G. Wolf, M. Eickenberg, **E. Belilovsky**. Channel-Selective Normalization for Label-Shift Robust Test-Time Adaptation. *Conference on Life Long Learning Agents (CoLLAs)*, Pisa, Italy, 2024.

(C11) G. Nanfack, A. Fulleringer*, J. Marty, M. Eickenberg, **E. Belilovsky**. Adversarial attacks on the interpretation of neuron activation maximization. *AAAI Conference on Artificial Intelligence (AAAI)*, Vancouver, Canada, 2024.

(C12) G. Legate*, N. Bernier*, L. Caccia, E. Oyallon, **E. Belilovsky**. Guiding The Last Layer in Federated Learning with Pre-Trained Models. *Advances in Neural Information Processing Systems (NeurIPS)*, New Orleans, USA, 2023.

(C13) Adel Nabli*, **E. Belilovsky**, E. Oyallon. A^2CiD^2: Accelerating Asynchronous Communication in Decentralized Deep Learning. *Advances in Neural Information Processing Systems (NeurIPS)*, New Orleans, USA, 2023.

(C14) N. Asadi*, M. Davari*, S. Mudur, R. Aljundi, **E. Belilovsky**. Prototype-Sample Relation Distillation: Towards Replay-Free Continual Learning. *International Conference on Machine Learning (ICML)*, Honolulu, USA, 2023.

(C15) L. Fournier*, S. Rivaud, **E. Belilovsky**, M. Eickenberg, E. Oyallon. Can Forward Gradient Match Backpropagation? *International Conference on Machine Learning (ICML)*, Honolulu, USA, 2023.

(C16) A. Sarfi*, Z. Karimpour*, N. Khalid*, M. Ravanelli, S. Mudur, **E. Belilovsky**. Simulated Annealing in Early Layers Leads to Better Generalization. *Conference on Computer Vision and Pattern Recognition (CVPR)*, Vancouver, Canada, 2023.

(C17) G. Legate*, L. Caccia*, **E. Belilovsky**. Re-Weighted Softmax Cross-Entropy to Control Forgetting in Federated Learning. *Second Conference on Lifelong Learning Agents (CoLLAs)*, Montreal, Canada , 2023.

(C18) M. Davari*, S. Horoi*, A. Natik*, G. Lajoie, G. Wolf, **E. Belilovsky**. Reliability of CKA as a Measure of Similarity in Deep Learning. *International Conference on Learning Representations (ICLR)*, Kigali, Rwanda, 2023.

(C19) N. Khalid*, T. Xie*, **E. Belilovsky**, T. Popa. CLIP-mesh: Generating textured meshes from text using pretrained image-text models. *SIGGRAPH Asia*, Daegu, South Korea, 2022.

(C20) M. Ernoult, F. Normandin*, A. Moudgil*, S. Spinney*, **E. Belilovsky**, I. Rish, B. Richards, Y. Bengio. Towards scaling difference target propagation by learning backprop targets. *International Conference on Machine Learning (ICML)*, Baltimore, USA, 2022.

(C21) M. Davari*, N. Asadi*, S. Mudur, R. Aljundi, **E. Belilovsky**. Probing Representation Forgetting in Supervised and Unsupervised Continual Learning. *Conference on Computer Vision and Pattern Recognition (CVPR)*, New Orleans, USA, 2022.

(C22) M. Yazdanpanah*, A. Abdul-Rahman*, M. Chaudhary*, C. Desrosiers, M. Havaei, **E. Belilovsky**, S. Ebrahimi Kahou. Revisiting Learnable Affines for Batch Norm in Few-Shot Transfer Learning. *Conference on Computer Vision and Pattern Recognition (CVPR)*, New Orleans, USA, 2022.

(C23) S. Gauthier*, B. Therien*, L. Alsène-Racicot*, M. Chaudhary*, **E. Belilovsky**[§], M. Eickenberg[§], G. Wolf[§]. Parametric Scattering Networks. *Conference on Computer Vision and Pattern Recognition (CVPR)*, New Orleans, USA, 2022. Oral presentation (acceptance rate <20%). [§]Equal contribution.

(C24) L. Caccia*, R. Aljundi, N. Asadi*, T. Tuytelaars, J. Pineau, **E. Belilovsky**. Reducing Representation Drift in Online Continual Learning. *International Conference on Learning Representations (ICLR)*, Virtual Conference, 2022.

(C25) L. Thiry*, M. Arble, **E. Belilovsky**, E. Oyallon. The Unreasonable Effectiveness of Patches in Deep Convolutional Kernel Methods. *International Conference on Learning Representations (ICLR)*, Virtual Conference, 2021.

(C26) B. Knyazev*, H. d. Vries, C. Cangea, G. W. Taylor, A. Courville, **E. Belilovsky**. Generative Compositional Augmentations for Scene Graph Prediction. *International Conference on Computer Vision (ICCV)*, Virtual Conference, 2021.

(C27) M. Michalkiewicz*, S. Parisot, S. Tsogkas, M. Baktashmotlagh, A. P. Eriksson, **E. Belilovsky**. Few-Shot Single-View 3-D Object Reconstruction with Compositional Priors. *European Conference on Computer Vision (ECCV)*, Virtual Conference, 2020.

(C28) B. Knyazev*, H. d. Vries, C. Cangea, G. W. Taylor, A. Courville, **E. Belilovsky**. Graph Density-Aware Losses for Novel Compositions in Scene Graph Generation. *British Machine Vision Conference (BMVC)*, Virtual Conference, 2020.

(C29) M. Michalkiewicz*, **E. Belilovsky**, M. Baktashmotlagh, A. P. Eriksson. A Simple and Scalable Shape Representation for 3D Reconstruction. *British Machine Vision Conference (BMVC)*, Virtual Conference, 2020.

(C30) L. Caccia*, **E. Belilovsky**, M. Caccia, J. Pineau. Online Learned Continual Compression with Adaptive Quantization Modules. *International Conference on Machine Learning (ICML)*, Virtual Conference, 2020.

(C31) **E. Belilovsky**, M. Eickenberg, E. Oyallon. Decoupled Greedy Learning of CNNs. *International Conference on Machine Learning (ICML)*, Virtual Conference, 2020.

(C32) R. Aljundi[§], L. Caccia[§], **E. Belilovsky**[§], M. Caccia[§], M. Lin, L. Charlin, T. Tuytelaars. Online Continual Learning with Maximally Interfered Retrieval. *Advances in Neural Information Processing Systems (NeurIPS)*, Vancouver, Canada, 2019. [§]Equal Contribution.

(C33) C. Cangea, **E. Belilovsky**, P. Lio, A. Courville. Bridging the Gap between Visual and Embodied Question Answering. *British Machine Vision Conference (BMVC)*, Cardiff, UK, 2019.

(C34) **E. Belilovsky**, M. Eickenberg, E. Oyallon. Greedy Layerwise Learning Can Scale to Imagenet. *International Conference on Machine Learning (ICML)*, Long Beach, USA, 2019.

(C35) E. Oyallon, **E. Belilovsky**, S. Zagoruyko, M. Valko. Compressing the Input for CNNs with the First-Order Scattering Transform. *European Conference on Computer Vision (ECCV)*, Munich, Germany, 2018.

(C36) **E. Belilovsky**, K. Kastner, G. Varoquaux, and M. B. Blaschko. Learning to Discover Sparse Graphical Models. *International Conference on Machine Learning (ICML)*, Sydney, Australia, 2017.

(C37) E. Oyallon, **E. Belilovsky**, and S. Zagoruyko. Scaling the Scattering Transform: Deep Hybrid Networks. *International Conference on Computer Vision (ICCV)*, Venice, Italy, 2017.

(C38) **E. Belilovsky**, G. Varoquaux, and M. B. Blaschko. Testing for Differences in Gaussian Graphical Models: Applications to Brain Connectivity. *Advances in Neural Information Processing Systems (NIPS)*, Barcelona, Spain, 2016. Oral (top 2.1% of submitted papers).

(C39) W. Bounliphone§, **E. Belilovsky**§, M. B. Blaschko, I. Antonoglou, A. Gretton. A Test of Relative Similarity for Model Selection in Generative Models. *International Conference on Learning Representations (ICLR)*, San Juan, Puerto Rico, 2016. §Equal Contribution.

Selected Workshop Papers and Preprints

(W1) A. M. Sarfi, B. Thérien, J. Lidin, **E. Belilovsky**. Communication-Efficient LLM Pretraining with SparseLoCo. *NeurIPS 2025 Workshop on Optimization for Machine Learning (OPT 2025)*.

(W2) B. Thérien, X. Huang, I. Rish, **E. Belilovsky**. MuLoCo: Muon is a Practical Inner Optimizer for DiLoCo. *ICML 2025 Workshop on Efficient Systems for Foundation Models (ES-FoMo III)*.

(W3) V. Singh, Z. Khalid, E. Oyallon, **E. Belilovsky**. Model Parallelism with Subnetwork Data Parallelism. *ICML 2025 Workshop on Efficient Systems for Foundation Models (ES-FoMo III)*.

(W4) A. Moudgil, B. Knyazev, **E. Belilovsky**. Towards a Learned-Optimization Free Lunch.

(W5) X. Huang, B. Thérien, **E. Belilovsky**. Towards Efficient Unroll Generalization in Learned Optimizers. *NeurIPS 2025 Workshop on Optimization for Machine Learning (OPT 2025)*.

(W6) S. Horoi, G. Wolf, **E. Belilovsky**, G. K. Dziugaite. Less is More: Undertraining Experts Improves Model Upcycling. *arXiv:2506.14126*, 2025.

(W7) M. R. Davari, U. Garg, W. Cai, **E. Belilovsky**. Rethinking Prompt Optimization: Reinforcement, Diversification, and Migration in Blackbox LLMs. *arXiv:2507.09839*, 2025.

(W8) P. Janson, B. Thérien, Q. Anthony, X. Huang, A. Moudgil, **E. Belilovsky**. PyLO: Towards Accessible Learned Optimizers in PyTorch. *ICML 2025 Workshop on Code for Machine Learning (CODEML)*, Vancouver, Canada, 2025. In submission to *MLSys* 2026.

(W9) B. Thérien*, C. Joseph*, B. Knyazev, E. Oyallon, I. Rish, **E. Belilovsky**. µLO: Compute-Efficient Meta-Generalization of Learned Optimizers. *NeurIPS Workshop on Optimization*, Vancouver, Canada, 2024. Oral. In Submission to ICLR 2026

(W10) H. Hameed*, G. Nanfack, **E. Belilovsky**. Not Only the Last-Layer Features for Spurious Correlations: All Layer Deep Feature Reweighting. *ECCV Workshop*, Milan, Italy, 2024.

(W11) P. Janson*, T. Popa, **E. Belilovsky**. Towards Motion from Video Diffusion Models. *ECCV Workshop on Foundation Models for 3D Humans*, Milan, Italy, 2024.

(W12) A. Zamani*, A. G. Aghdam, T. Popa, **E. Belilovsky**. Temporally Consistent Object Editing in Videos using Extended Attention. *CVPR Workshop*, Seattle, United States, 2024.

(W13) A. Moudgil*, B. Knyazev, G. Lajoie, **E. Belilovsky**. Learning to Optimize with Recurrent Hierarchical Transformers. *ICML Workshop*, Honolulu, United States, 2023

(W14) M. Sonwa*, J. Hansen, **E. Belilovsky**. Imitation from Observation with Bootstrapped Contrastive Learning. *NeurIPS Offline Reinforcement Learning Workshop*, New Orleans, United States, 2022.

(W15) A. Patel*, M. Eickenberg, **E. Belilovsky**. Local Learning with Neuron Groups. *ICLR Workshop from Cells to Societies*, Virtual, 2022.

(W16) N. Asadi*, S. Mudur, **E. Belilovsky**. Tackling Online One-Class Incremental Learning by Removing Negative Contrasts. *NeurIPS Workshop on Distribution Shifts*, Virtual, 2021.

(W17) A. Anand, **E. Belilovsky**, K. Kastner, H. Larochelle, A. Courville. Blindfold Baselines for EmbodiedQA. *NeurIPS VIGIL Workshop*, Montréal, Canada, 2018.

(W18) **E. Belilovsky**, M.B. Blaschko, J.R. Kiros, R. Urtasun, R. Zemel. Joint Embeddings of Scene Graphs and Images. *ICLR Workshop Track*, Toulon, France, 2017.